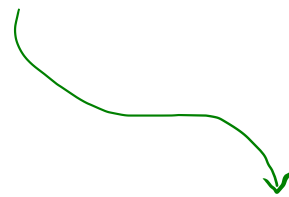


fram / آئندہ میں ^{ہے} جان



$$\underline{y = wx + b}$$



بدلت آمدن w, b
[کچھ متغیرات کمتر سے متعلق ہوں!]

min



$\begin{bmatrix} w \\ b \end{bmatrix}$



min (loss funct)



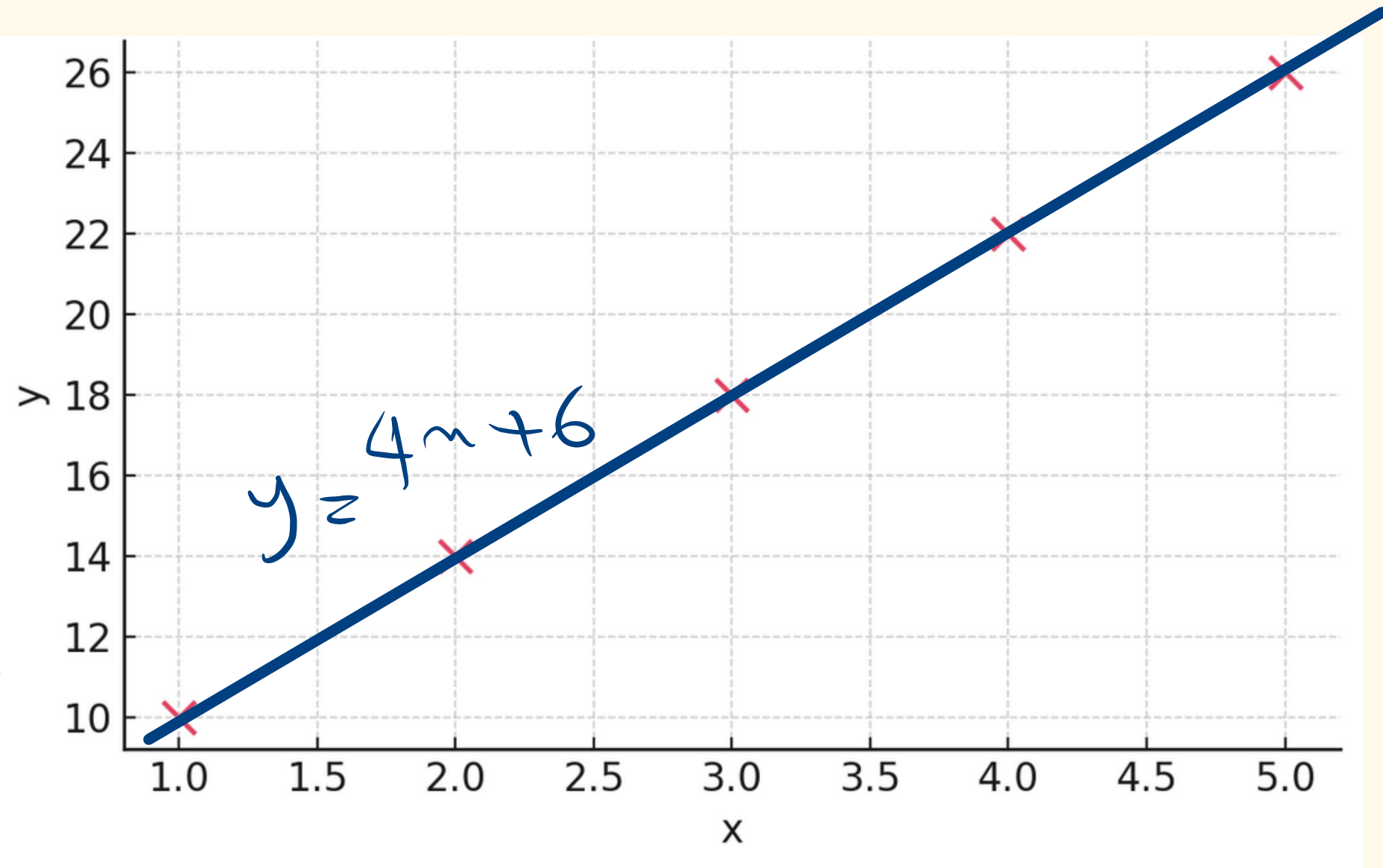
	مساحت خانه (X1)	قیمت خانه (Y)
خانه ۱	1	10
خانه ۲	2	14
خانه ۳	3	18
خانه ۴	4	22
خانه ۵	5	26

$$y = 4x + 6$$

رابطه ی ویژگی ها و لیبل
را به دست بیاورید؟



x	y
1	10
2	14
3	18
4	22
5	26





1	10
2	14
3	18
4	22
5	26

به راحتی به دست می آید :

$$y = \overset{4}{3}x + \overset{6}{7}$$

اما اگر کمی رابطه مشکل تر شود چطور؟



$$y = 4x + 6 \rightarrow$$

x y
1 → 10

1	10
2	14
3	18
4	22
5	26

به راحتی به دست می آید :

$$y = \overset{4}{\cancel{3}} \hat{x} + \overset{6}{\cancel{7}} \rightarrow \text{loss fun: } 0$$

وزن (w) بایاس (b)

اما اگر کمی رابطه مشکل تر شود چطور؟



$$y = 4x + 6$$

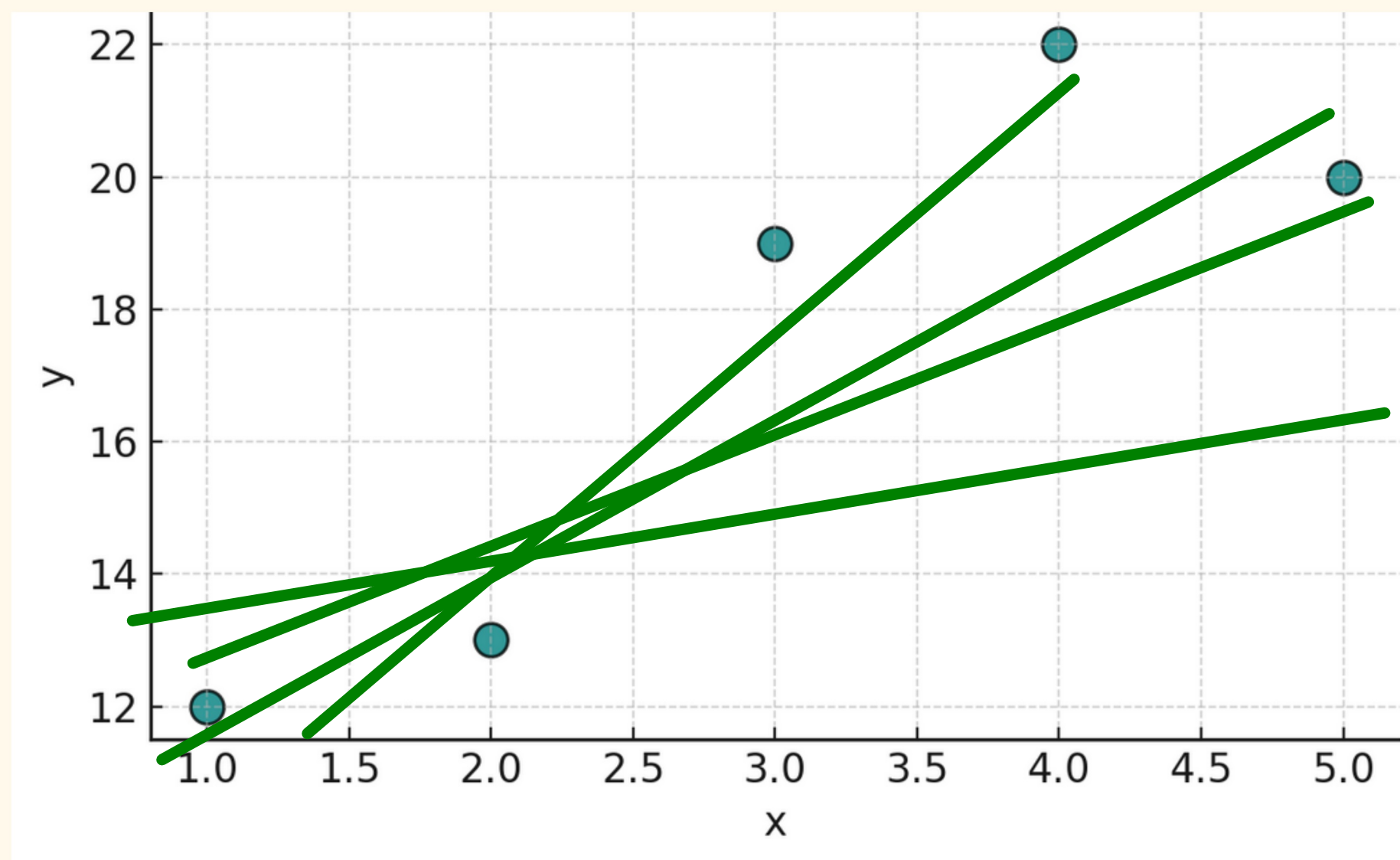
x y

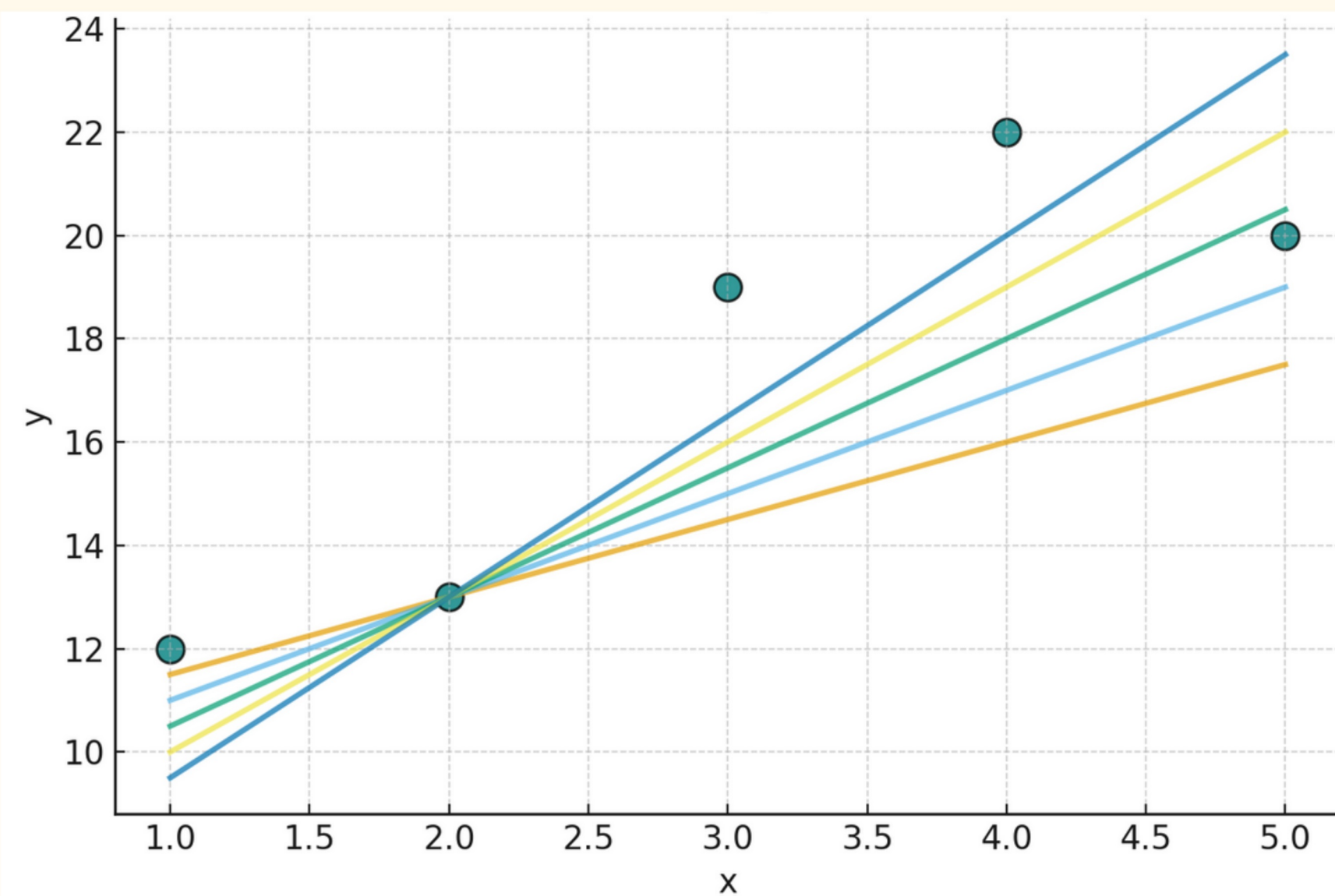
1	12
2	13
3	19
4	22
5	20

الان چطور؟



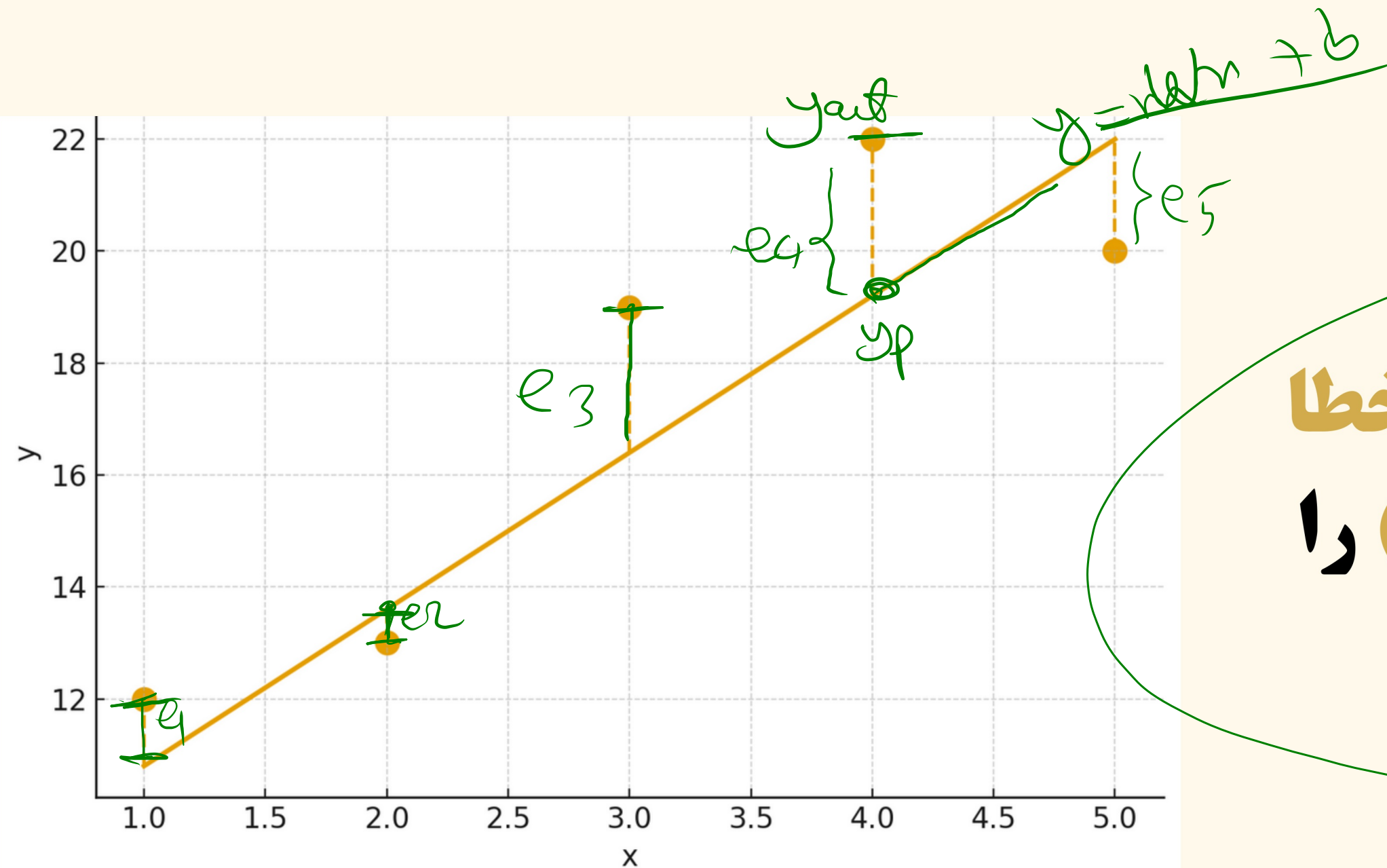
1	12
2	13
3	19
4	22
5	20





کدام خط می تواند
جواب باشد؟

$w, b \rightarrow \min(MSE)$



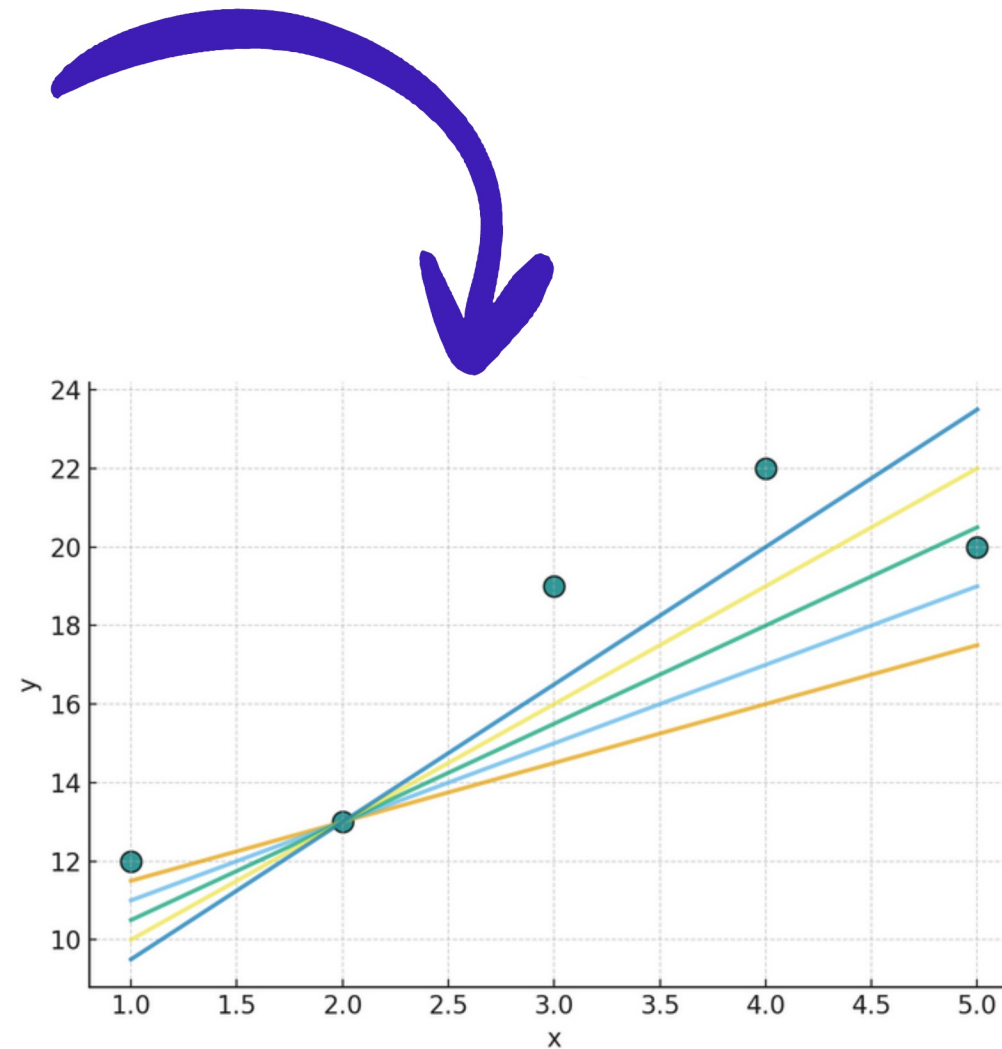
نموداری که کمترین خطا
(یا loss function) را

دارد!

MSE



1	12
2	13
3	19
4	22
5	20



خطی بهترین نماینده (پیش
بینی کننده) برای داده مورد
نظر است که به کمترین
MSE منجر شود.

$$\text{loss func (mse)} = \frac{1}{n} \sum_{i=1}^n (\overset{\text{actual}}{y_i} - \overset{\text{predicted: } wX+b}{\hat{y}_i})^2$$

یادگیری ماشین (مقدمه)

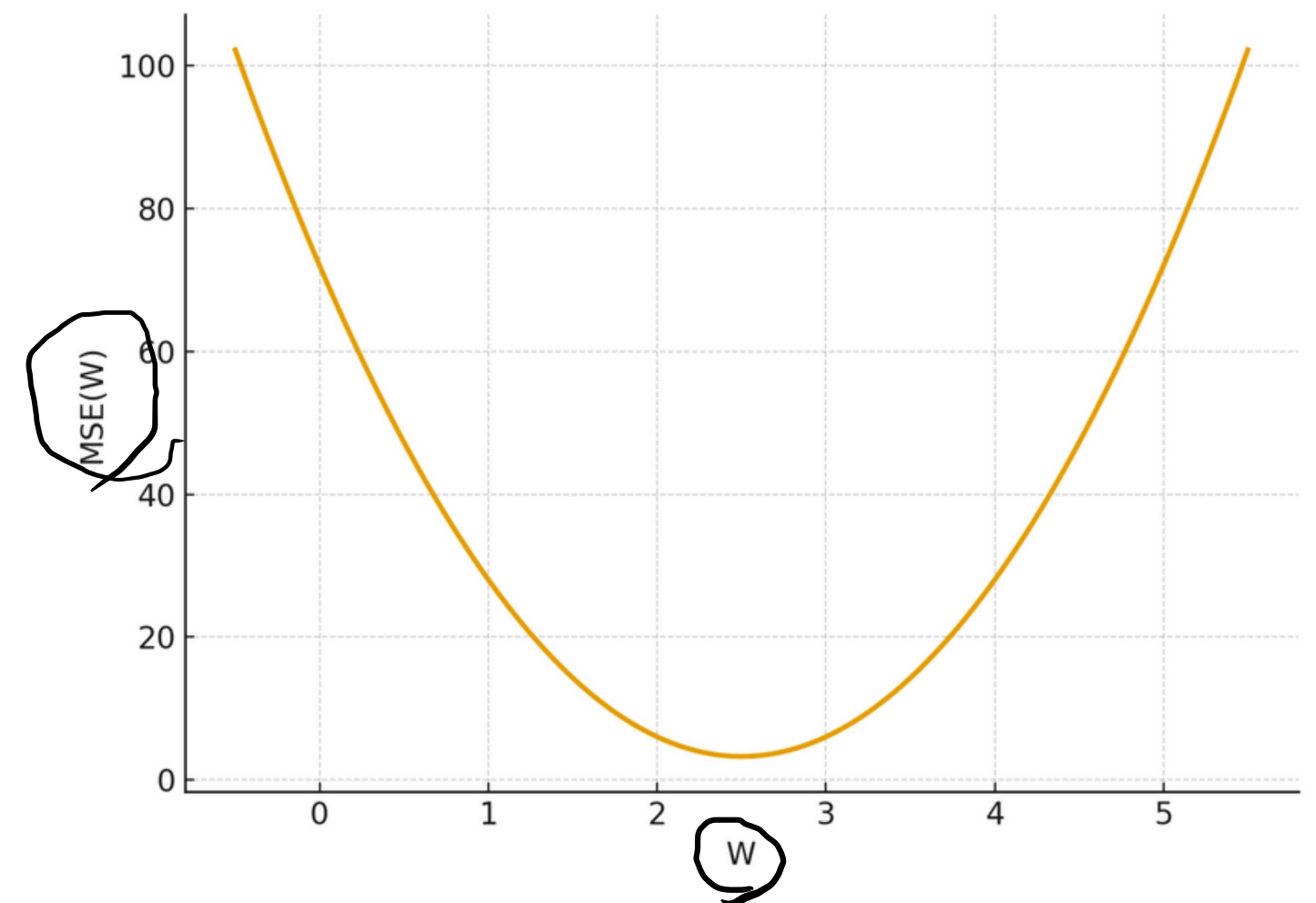


$$\underline{w, b} \rightarrow \min (\text{mse})$$

$$\text{mse} = \frac{1}{n} \sum_{i=1}^n (y_i - (wx_i + b))^2$$

چگونه به سمت
loss func(MSE)
مینیمم حرکت کنیم؟

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

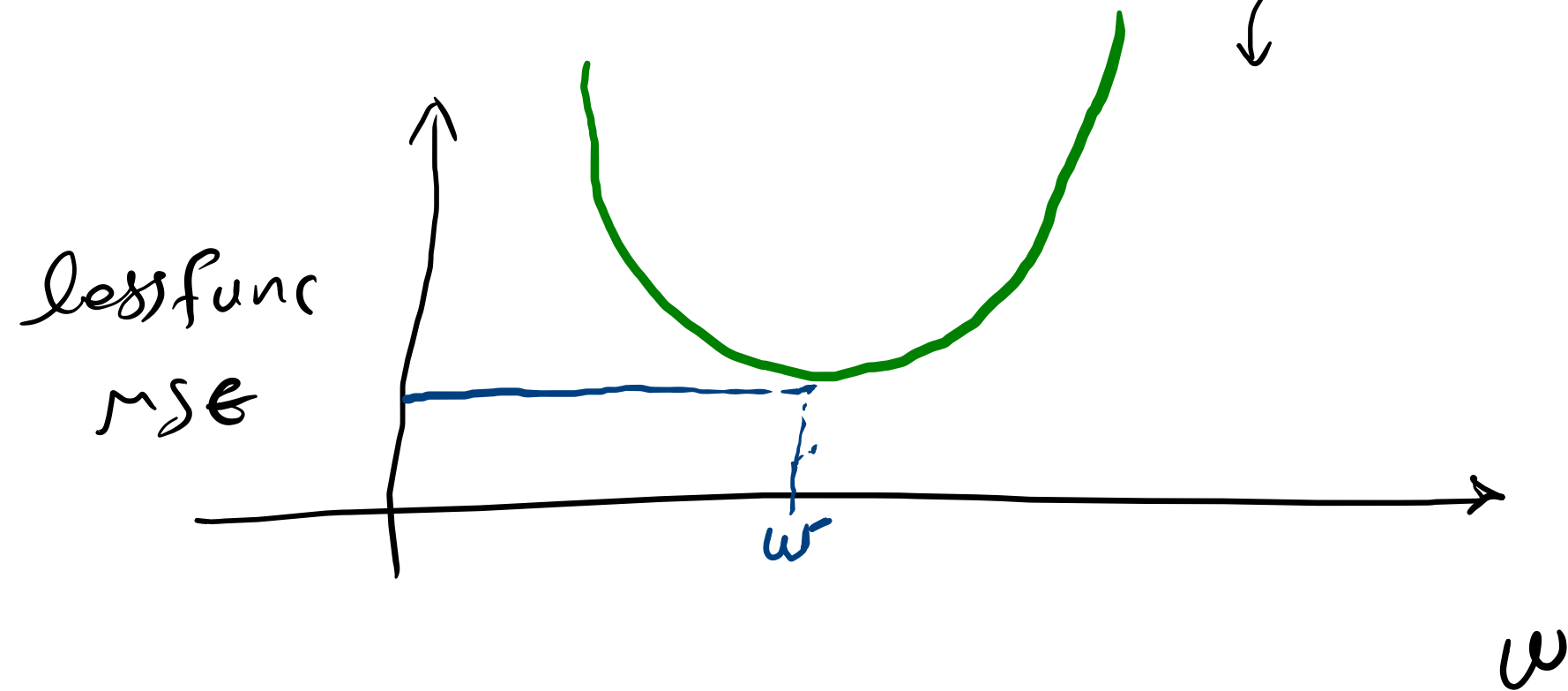


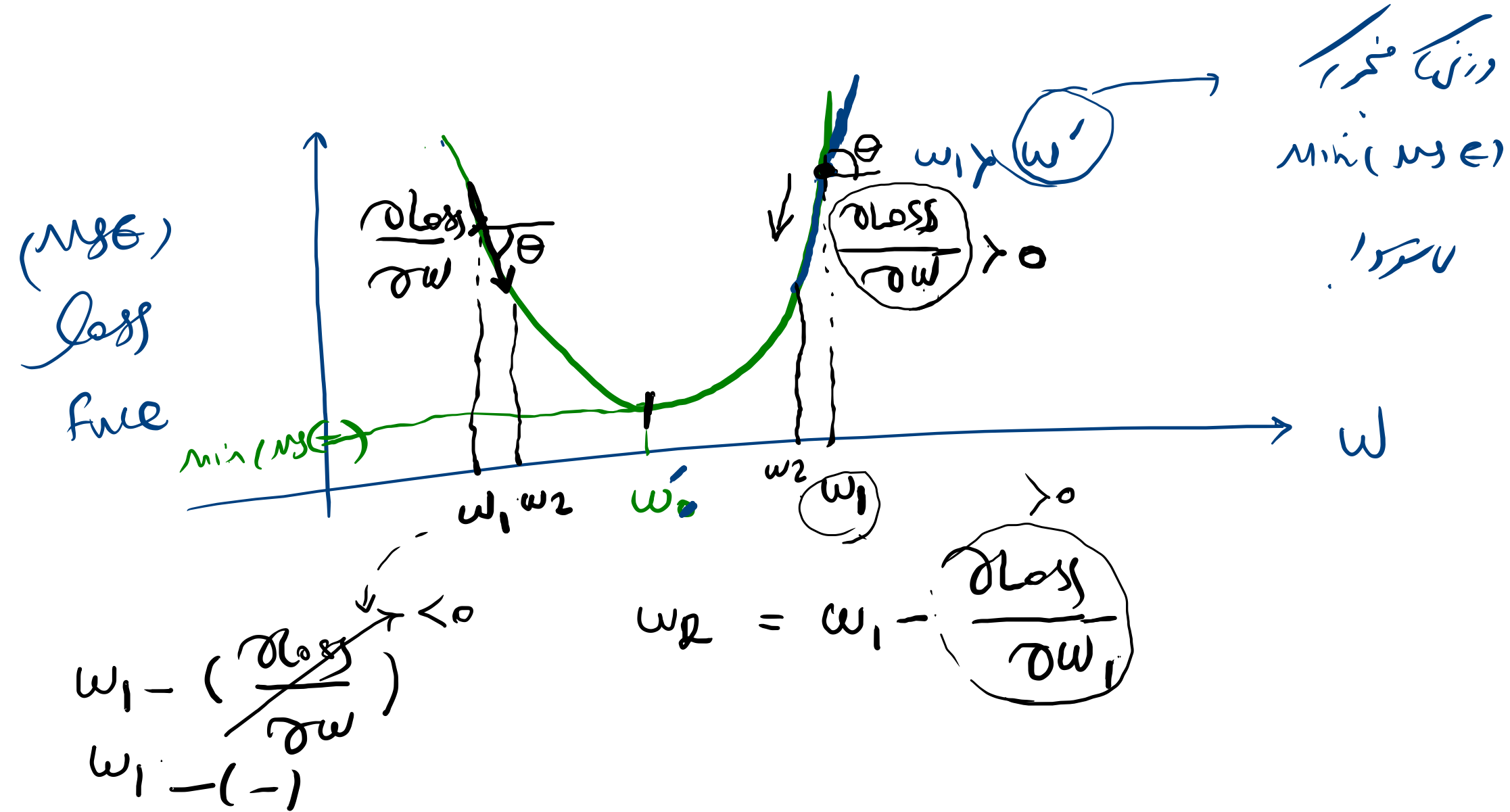
$$MSE = \frac{1}{n} \sum_{i=1}^n \left(\underbrace{y_i - (wx_i + b)}_{\text{residual}} \right)^2 \rightarrow$$

$$y_i^2 - 2(y_i (wx_i)) + \underbrace{(wx_i)^2}$$

$$MSE \propto w^2$$

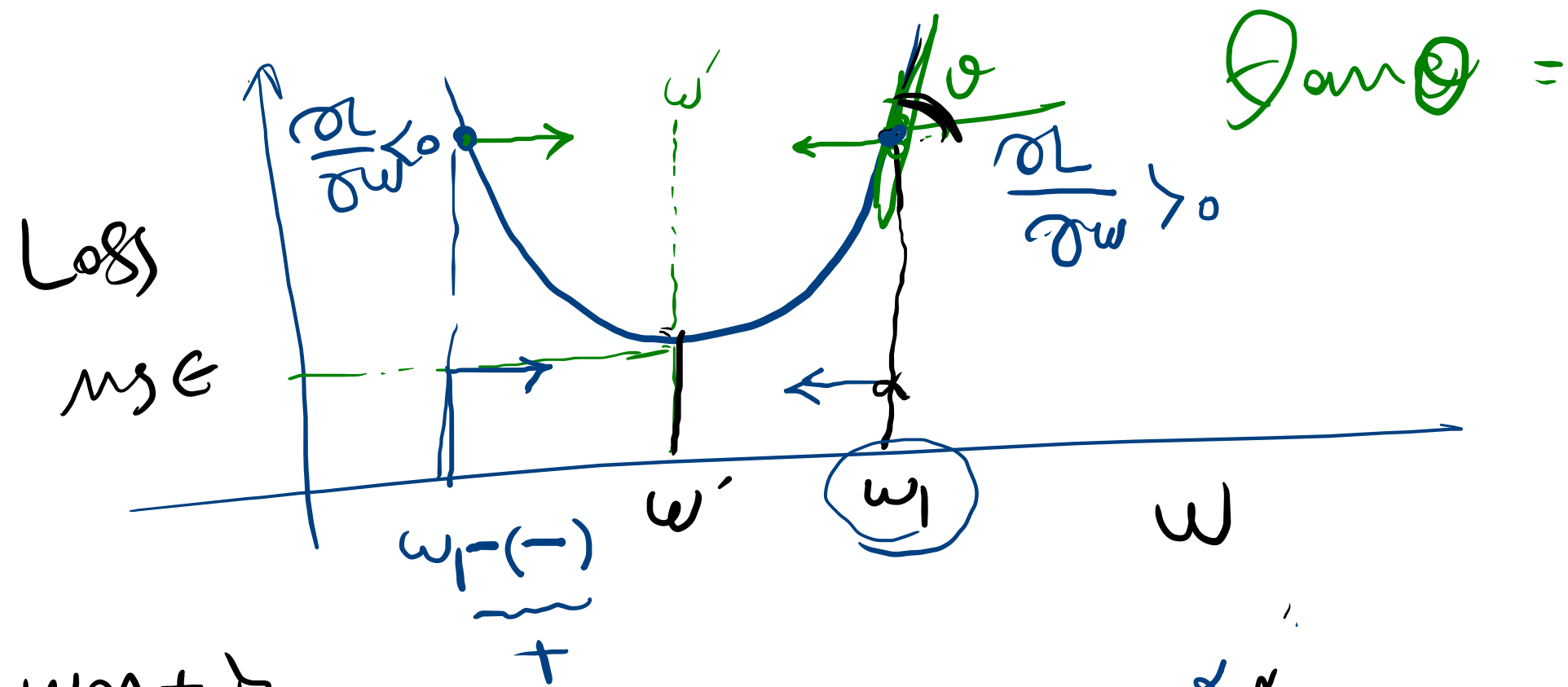
$$MSE \propto w^2$$





Randomly مقدار w به صورت تصادفی!

$$w_2 = w_1 - \frac{\partial \text{loss}}{\partial w_1}$$



$$y = \frac{w}{\tau} + b$$

$$w_2 = w_1 - \left(\frac{\partial \text{Log}}{\partial w} \right) \alpha$$

$$\text{Derivative} =$$

$$\frac{\Delta y}{\Delta x}$$

$$= \Delta \frac{\partial y}{\partial x}$$

$$\frac{\partial \text{Log}}{\partial w}$$

x	y
1	5
2	7

SLR = Single Var. Lin. Reg.

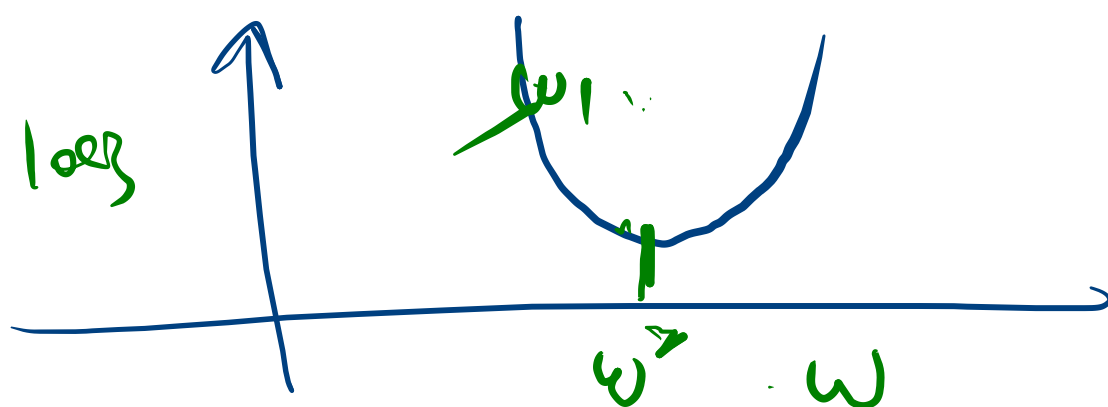
$$\hat{y} = \underbrace{w}_{\text{weight}} \underbrace{x}_{\text{feature}} + \underbrace{b}_{\text{bias}}$$

$$(w, b) \rightarrow \min(MSE) = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$\hat{y}_i = wx_i + b$

act pred

$$MSE = \frac{1}{n} \sum (y_i - (wx_i + b))^2 \rightarrow \boxed{MSE \propto w^2}$$



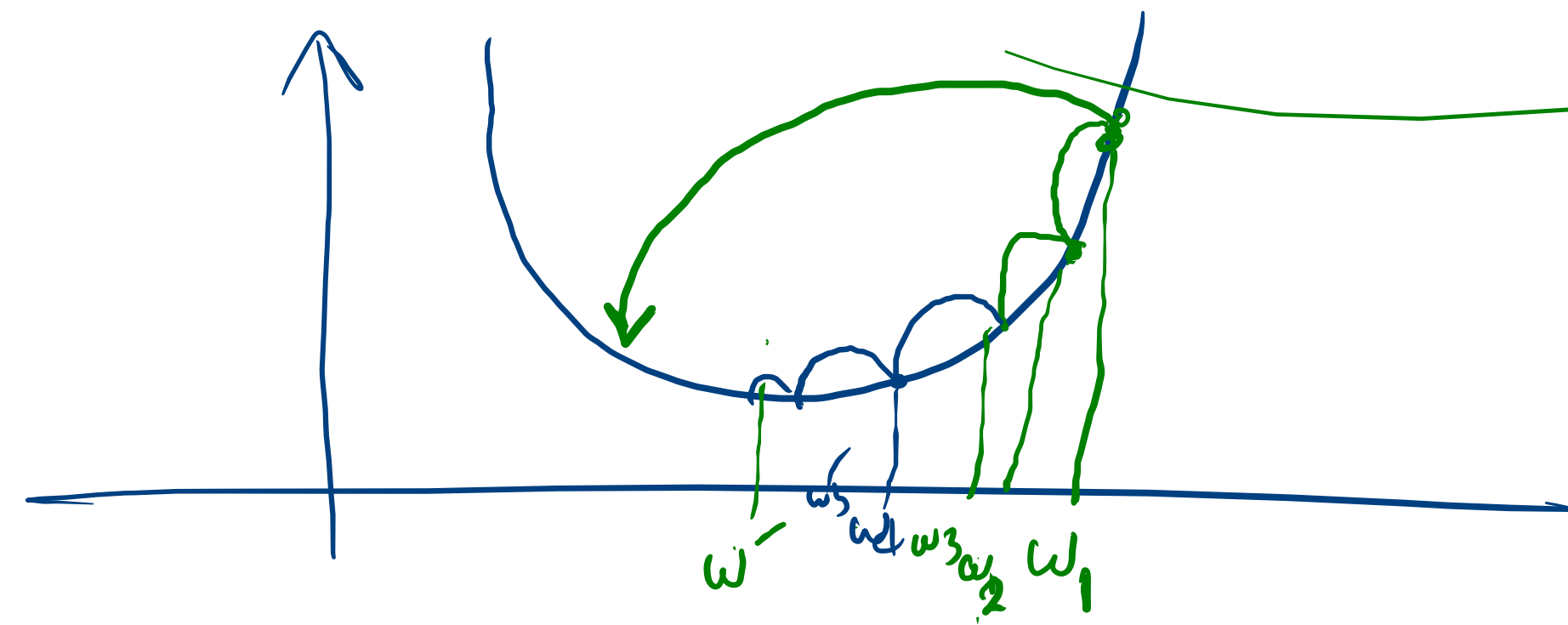
MSE

$y = wx + b \rightarrow w \rightarrow \min(MSE)$

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w}$$

(Note: In the original image, 'L' is circled in blue and labeled 'MSE' with an arrow, and 'w' is underlined in the denominator.)

gradient



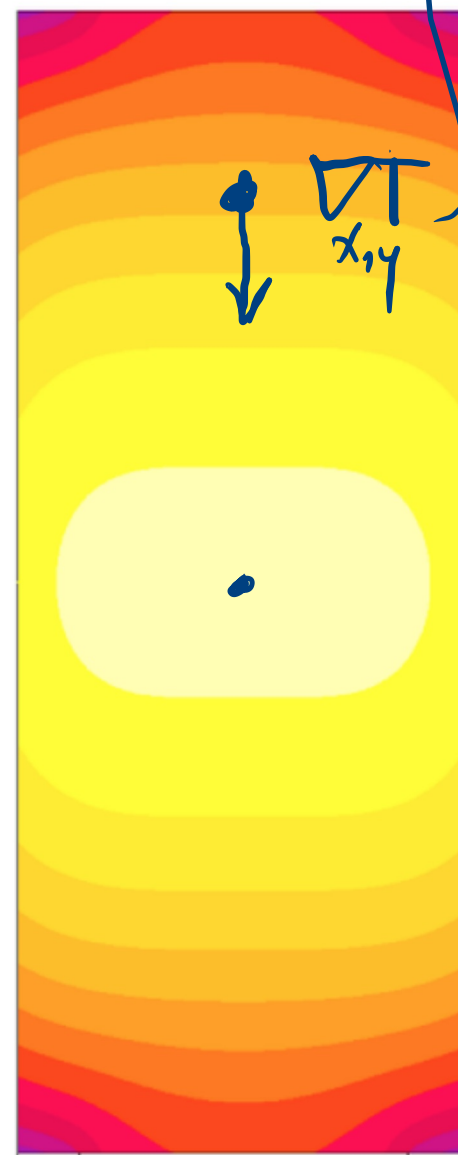
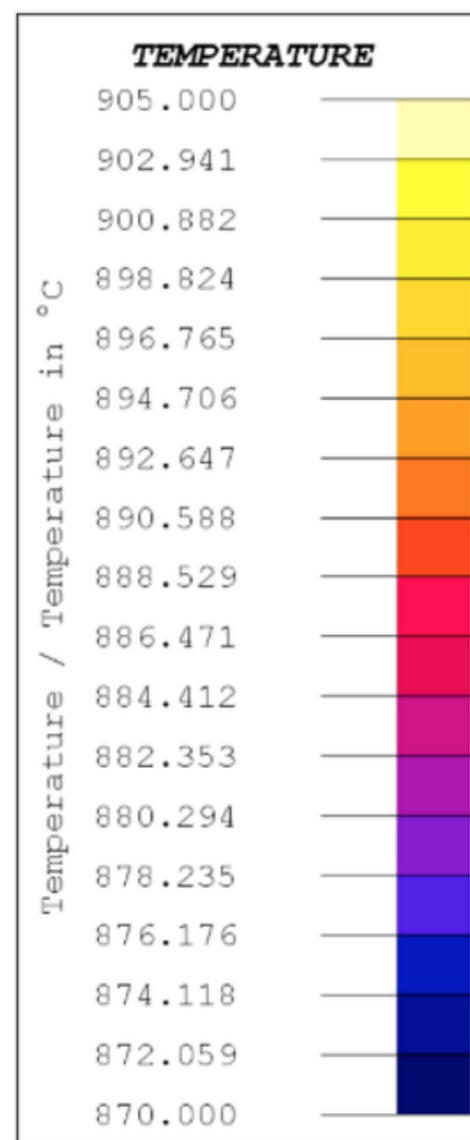
در هر بار به جلو

$$\nabla L = \frac{\partial L}{\partial w}$$

→ -جواب
max

$$\nabla T = \frac{\partial T}{\partial x} \hat{i} + \frac{\partial T}{\partial y} \hat{j}$$

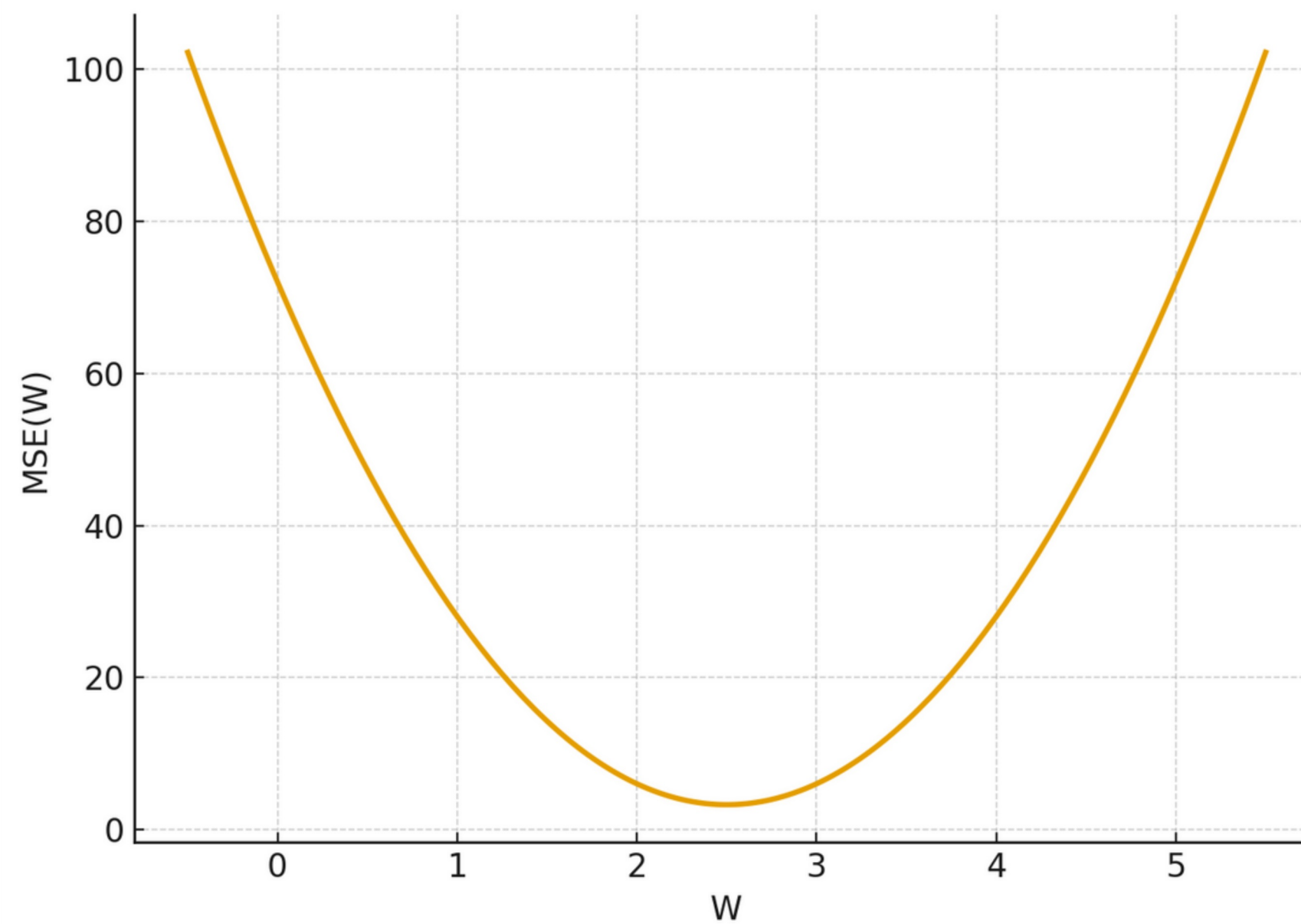
یادگیری ماشین (مقدمه)



$-\nabla T$



$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



$$W_{new} = W_{old} - \alpha \frac{\partial L}{\partial W}$$



$$*W_x = W_x - \alpha \left(\frac{\partial \text{Error}}{\partial W_x} \right)$$

Diagram illustrating the weight update formula with labels:

- New weight**: Points to $*W_x$
- Old weight**: Points to W_x
- Learning rate**: Points to α
- Derivative of Error with respect to weight**: Points to $\left(\frac{\partial \text{Error}}{\partial W_x} \right)$

input X	target Y
→ 1	2
→ 3	6

$$y = \underbrace{2x}_p$$

مزمنہ کرنے سے ہاتھ آئے!

$$y = wx + b$$

gradient descent

$$w_{new} = w_{old} - \underbrace{\alpha}_{\text{ا.ر.}} \frac{\partial L}{\partial w_{old}}$$

ا. ماہر، عدد، رقم $w = 1$ قرار دے رہے!

mse (loss): $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

$$Loss = \frac{1}{n} \sum (y_i - (wx_i + b))^2 \Rightarrow$$

$$w_{new} = w_{old} - \alpha \left(\frac{\partial}{\partial w} \left[\frac{1}{n} \sum (y_i - (wx_i + b))^2 \right] \right)$$

$x^h = nm' a^{h-1}$

$$F = \frac{1}{n} \frac{\partial}{\partial w} (y_i - (wx_i + b))^2 = \frac{1}{n} (2) (-x_i) (y_i - (wx_i + b))$$

F ←

$$F = -\frac{2}{n} x_i (y_i - (wx_i + b))$$

$$w_{new} = w_{old} - \eta F$$

$$F = -\frac{2}{n} \sum (x_i) (y_i - (w_{old} x_i + b))$$

	X	y
x_1	1	2 y_1
x_2	3	6 y_2

$$F = -\frac{2}{2} \left[1 (2 - [1 (1) + 0]) \right] + \left[3 (6 - (1 \times 3)) \right]$$

$$F = -1 (1 + 9) = -10$$

$$w_{new} = w_0 (1 - \alpha (F))$$

↓

$$1 - (0.1)(-1.0)$$

$$w_{new} = 1 + 0.1 = 1.1$$

↓

$$w_0 1d$$



$$w_{new} = w_0 (1 - \alpha \frac{\partial L}{\partial w})$$

$$y = w^T x$$

↓

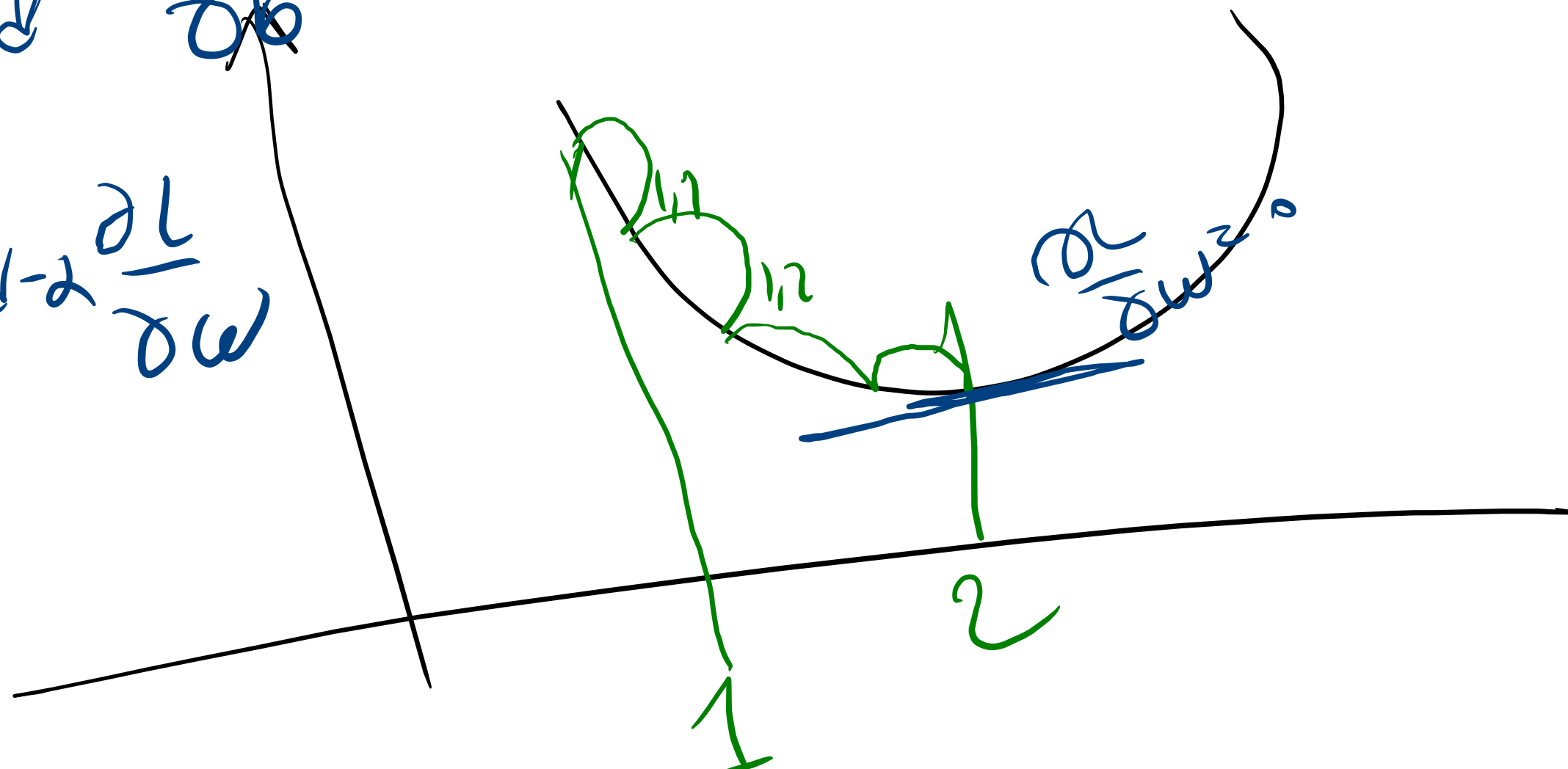
$$1$$

" "

$$y = 2x$$

$$b_{new} = b_{old} - \frac{\partial L}{\partial b}$$

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w}$$

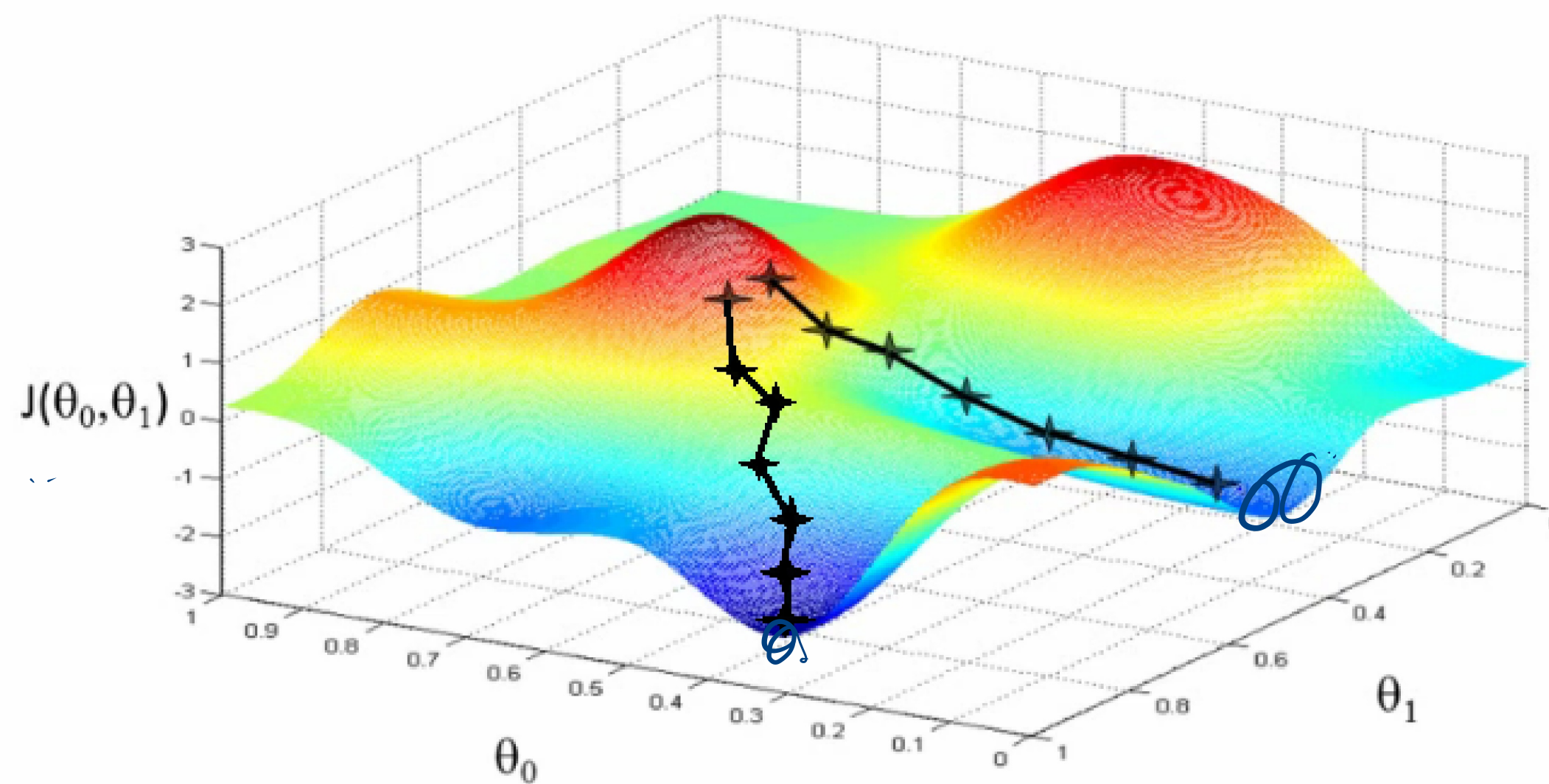


$$w_{new} = w_{old}$$

or

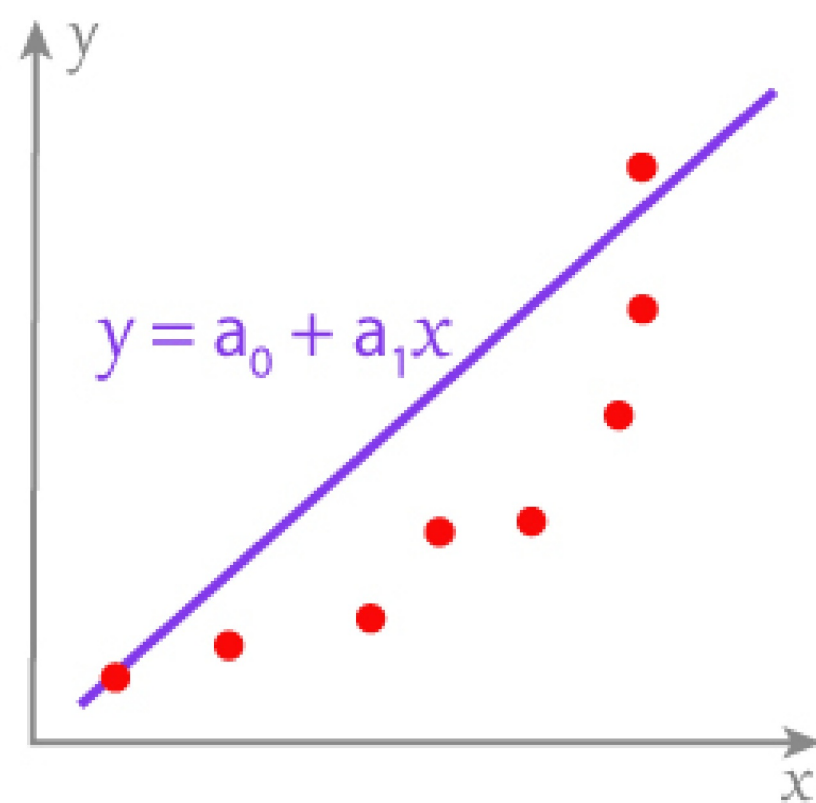


لایه

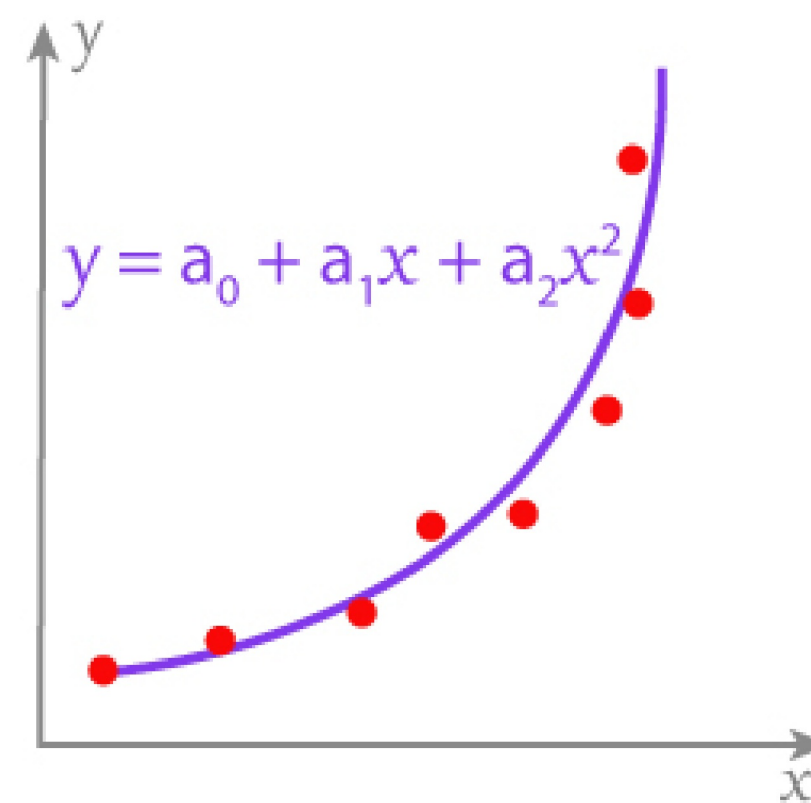




Simple Linear Progression



Polynomial Progression





Simple Linear
Regression

$$y = b_0 + b_1 x_1$$

Multiple Linear
Regression

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

Polynomial Linear
Regression

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + \dots + b_n x_n^n$$